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GitHub link: https://github.com/taniyasif/Python-labs

LAB: 01

Task 1:

Make 2-2 programs of each datatype.

- **NUMERIC TYPES:**

❖ **Integer:**

```
num1 = 4
num2 = 8
sum = num1 + num2
print("The sum is:", sum)
```

```
The sum is: 12
```

```
num = 2

if num % 2 == 0:
    print("The number is even")
else:
    print("The number is odd")
```

```
The number is even
```

❖ Float:

```
num1 = 3.5
num2 = 8.5
result = num1 - num2

print("The result is:", result)
```

```
The result is: -5.0
```

```
num1 = 1.0
num2 = 5.0

result = num1 / num2

print("The answer is:", result)
```

```
The answer is: 0.2
```

❖ Complex:

```
num1 = 4 + 4j
num2 = 5 + 4j
result = num1 + num2
print("The sum is:", result)
```

```
The sum is: (9+8j)
```

```
num1 = 6 + 3j
num2 = 2 + 2j

result = num1 * num2

print("The product is:", result)
```

```
The product is: (6+18j)
```

- **SEQUENCE TYPES:**

❖ String

```
first_name = "Aqsa"
last_name = "Ilyas"
full_name = first_name + " " + last_name
print("Full name is:", full_name)
```

```
Full name is: Aqsa Ilyas
```

```
message = "Hello, world!"
print(message)
```

```
Hello, world!
```

❖ List:

```
fruits = ["orange", "banana", "watermelon"]
for fruit in fruits:
    print(fruit)
```

```
orange
banana
watermelon
```

```
numbers = [1, 2, 3]
numbers.append(5)
print("Updated list:", numbers)
```

```
Updated list: [1, 2, 3, 5]
```

❖ Tuple:

```
]: colors = ("pink", "brown", "blue")
   print("Second color is:", colors[1])
```

```
Second color is: brown
```

```
]: 
   fruits = ("grapes", "banana", "cherry")

   for fruit in fruits:
       print(fruit)
```

```
grapes
banana
cherry
```

❖ Range:

```
11]: for num in range(1, 6):
         print(num)
```

```
1
2
3
4
5
```

```
12]: for num in range(2, 11, 2):
         print(num)
```

```
2
4
6
8
10
```

● SET TYPES:

❖ Set:

```python
fruits = {"apple", "banana", "mango"}

print("Fruits set:", fruits)
```

```
Fruits set: {'apple', 'banana', 'mango'}
```

```python
numbers = {1, 2, 3}
numbers.add(4)

print("Updated set:", numbers)
```

```
Updated set: {1, 2, 3, 4}
```

❖ Frozen set:

```python
5]: num = frozenset([1,2,3,4])

    print("num frozenset:", num)
```

```
num frozenset: frozenset({1, 2, 3, 4})
```

```python
6]: set1 = frozenset([1, 2, 3])
    set2 = frozenset([3, 4, 5])
    common = set1.intersection(set2)

    print("Common items:", common)
```

```
Common items: frozenset({3})
```

● MAPPING TYPE:

❖ Dictionary dict:

```python
student = {
    "name": "aqsa",
    "age": 18,
    "class": "BS IT"
}

print("Student Info:", student)
```

```
Student Info: {'name': 'aqsa', 'age': 18, 'class': 'BS IT'}
```

```python
person = {
    "name": "aqsa",
    "city": "Karachi"
}

print("Name is:", person["name"])
```

```
Name is: aqsa
```

- **BOLEAN TYPE:**

]:
```python
a = 30
b = 10

result = a > b
print("Is a greater than b?", result)
```

```
Is a greater than b? True
```

]:
```python
x = 10
y = 10

print("Are x and y equal?", x == y)
```

```
Are x and y equal? True
```

Task 2:

Make up to 5 shapes programs using *

```python
print("Square Shape:")
print("* * * * *")
print("* * * * *")
print("* * * * *")
print("* * * * *")
print("* * * * *")
print("Right-Angled Triangle:")
print("*")
print("* *")
print("* * *")
print("* * * *")
print("* * * * *")
print("Inverted Triangle:")
print("* * * * *")
print("* * * *")
print("* * *")
print("* *")
print("*")
```

```
Square Shape:
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
Right-Angled Triangle:
*
* *
* * *
* * * *
* * * * *
Inverted Triangle:
* * * * *
* * * *
* * *
* *
*
```

```python
print("Pyramid Shape:")
print("    *")
print("   * *")
print("  * * *")
print(" * * * *")
print("* * * * *")
print("Diamond Shape:")
print("    *")
print("   * *")
print("  * * *")
print("   * *")
print("    *")
```

```
Pyramid Shape:
    *
   * *
  * * *
 * * * *
* * * * *
Diamond Shape:
    *
   * *
  * * *
   * *
    *
```

]:

Task 3:

Make same shapes you have made in task 2, using * multiple by number.

Program:

]:

```
print("Square Shape:")
print("* " * 5)
print("* " * 5)
print("* " * 5)
print("* " * 5)
print("* " * 5)
print("Right-Angled Triangle:")
print("* " * 1)
print("* " * 2)
print("* " * 3)
print("* " * 4)
print("* " * 5)
print("Inverted Triangle:")
print("* " * 5)
print("* " * 4)
print("* " * 3)
print("* " * 2)
print("* " * 1)
print("Pyramid Shape:")
print(" " * 4 + "* ")
print(" " * 3 + "* " * 2)
print(" " * 2 + "* " * 3)
print(" " * 1 + "* " * 4)
print(" " * 0 + "* " * 5)
```

```
print("Diamond Shape:")
print(" " * 4 + "* ")
print(" " * 3 + "* " * 2)
print(" " * 2 + "* " * 3)
print(" " * 3 + "* " * 2)
print(" " * 4 + "* ")
```

```
Square Shape:
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
Right-Angled Triangle:
*
* *
* * *
* * * *
* * * * *
Inverted Triangle:
* * * * *
* * * *
* * *
* *
*
```

```
Pyramid Shape:
    *
   * *
  * * *
 * * * *
* * * * *
Diamond Shape:
    *
   * *
  * * *
   * *
    *
```

Lab 2:

Task 1:

Print numbers from 1 to 10 using a for loop.

Program:

```
[1]: for num in range(1, 11):
         print(num)
```

```
1
2
3
4
5
6
7
8
9
10
```

Task 2:

Print all even numbers between 1 and 20 using a while loop.

Program:

[2]:
```python
num = 2
while num <= 20:
    print(num)
    num += 2
```

```
2
4
6
8
10
12
14
16
18
20
```

Task 3:

Calculate the sum of numbers from 1 to 100 using a loop.

Program:

[3]:
```python
total = 0
for num in range(1, 101):
    total += num

print(f"The sum of numbers from 1 to 100 is: {total}")
```

```
The sum of numbers from 1 to 100 is: 5050
```

Task 4:

Print the multiplication table of 5 using a loop:

Program:

```
[4]: num = 5
     print(f"Multiplication Table of {num}:")

     for i in range(1, 11):
         print(f"{num} x {i} = {num * i}")
```

```
Multiplication Table of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Task 5:

Find the factorial of a given number using a for loop.

Program:

```
[5]: num = int(input("Enter a number to find its factorial: "))
factorial = 1
if num < 0:
    print("Factorial does not exist for negative numbers.")
else:
    for i in range(1, num + 1):
        factorial *= i

    print(f"The factorial of {num} is: {factorial}")
```

```
Enter a number to find its factorial:  4
The factorial of 4 is: 24
```

Task 6:

Create a list of numbers and print only the odd numbers using a loop.

Program:

```
[6]: numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
for num in numbers:
    if num % 2 != 0:
        print(num)
```

```
1
3
5
7
9
```

Task 7:

Iterate over a list of fruits and print each item.

Program:

```
fruits = ["apple", "banana", "cherry", "date", "elderberry", "fig", "grape"]
for fruit in fruits:
    print(fruit)
```

```
apple
banana
cherry
date
elderberry
fig
grape
```

LAB 4:

Task 1:

Write a python program to take 2 numbers as input and perform all arithmetic operations on them.

Program:

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
print("Addition:", num1 + num2)
print("Subtraction:", num1 - num2)
print("Multiplication:", num1 * num2)
print("Division:", num1 / num2)
print("Modulus (Remainder):", num1 % num2)
print("Exponent (Power):", num1 ** num2)
print("Floor Division:", num1 // num2)
```

```
Enter first number:  2
Enter second number:  7
Addition: 9.0
Subtraction: -5.0
Multiplication: 14.0
Division: 0.2857142857142857
Modulus (Remainder): 2.0
```

Task 2:

Create a function that takes two numbers and return their sum, difference, product, and quotient.

Program:

```
def calculate(a, b):
    sum_result = a + b
    difference = a - b
    product = a * b
    if b != 0:
        quotient = a / b
    else:
        quotient = "Undefined (division by zero)"

    return sum_result, difference, product, quotient
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
sum_result, difference, product, quotient = calculate(num1, num2)
print("Sum:", sum_result)
print("Difference:", difference)
print("Product:", product)
print("Quotient:", quotient)
```

```
Enter first number:  6
Enter second number:  9
Sum: 15.0
Difference: -3.0
Product: 54.0
Quotient: 0.6666666666666666
```

Task 3:

Write a python script to find the remainder when one number is divided by another.

Program:

```
num1 = int(input("Enter the first number: "))
num2 = int(input("Enter the second number: "))
remainder = num1 % num2
print("The remainder is:", remainder)
```

```
Enter the first number:  4
Enter the second number:  9
The remainder is: 4
```

Task 4:

Write a program to calculate the area of a circle using the formula: area=pi*r^2.

Program:

```python
import math
r = float(input("Enter the radius of the circle: "))
area = math.pi * r ** 2

# Showing the result
print("Area of the circle is:", area)
```

```
Enter the radius of the circle:  7
Area of the circle is: 153.93804002589985
```

Task 05:

Implement a program that takes a number as input and returns its square and cube using exponentiation.

Program:

```python
num = float(input("Enter a number: "))
square = num ** 2
cube = num ** 3
print("Square of the number is:", square)
print("Cube of the number is:", cube)
```

```
Enter a number:  5
Square of the number is: 25.0
Cube of the number is: 125.0
```

```python
print("Select operation:")
```

Task 6:

Create a simple calculator in python that allows the user to choose an operation(addition, subtraction, etc.)and inputs two numbers.

Program:

```
]: print("Select operation:")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")
choice = input("Enter choice (1/2/3/4): ")
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
if choice == '1':
    print("Result:", num1 + num2)
elif choice == '2':
    print("Result:", num1 - num2)
elif choice == '3':
    print("Result:", num1 * num2)
elif choice == '4':
    if num2 != 0:
        print("Result:", num1 / num2)
    else:
        print("Error: Cannot divide by zero.")
else:
    print("Invalid choice.")
```


```
else:
    print("Invalid choice.")
```


```
Select operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter choice (1/2/3/4):  3
Enter first number:  8
Enter second number:  6
Result: 48.0
```

Lab 5:

Task 1:

Basic Task: Write a program that checks if a given number is positive, negative, or zero.

Program:

```
number = float(input("Enter a number: "))

if number > 0:
    print("The number is positive.")
elif number < 0:
    print("The number is negative.")
else:
    print("The number is zero.")
```

```
Enter a number:  2
The number is positive.
```

Task 2:

Intermediate Task: Write a program that takes user input and determines whether it's a even or odd.

Program:

```
]: number = int(input("Enter a number: "))

if number % 2 == 0:
    print("The number is even.")
else:
    print("The number is odd.")
```

```
Enter a number:  7
The number is odd.
```

Task 3:

Advanced Task: Create a program that asks user to print:
"Excellent" if marks are above 80
"Good" if marks are between 60 and 80
"Needs Improvement" if marks are below 60

Program:

```
|: marks = float(input("Enter your marks: "))

if marks > 80:
    print("Excellent")
elif 60 <= marks <= 80:
    print("Good")
else:
    print("Needs Improvement")
```

```
Enter your marks:  55
Needs Improvement
```

Lab 6:

Task 1:

Basic Task: Write a for loop to print the first 10 natural numbers.

Program:

```
for i in range(1, 11):
    print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

Task 2:

Intermediate Task: Write a while loop that prints numbers from 10 down to 1

Program:

```
i = 10
while i >= 1:
    print(i)
    i -= 1
```

```
10
9
8
7
6
5
4
3
2
1
```

Task 3:

Advanced Task: Create a program that uses a for loop to iterate over a string and count the number of vowels.

Program:

```
text = input("Enter a string: ")
vowels = "aeiouAEIOU"
count = 0

for char in text:
    if char in vowels:
        count += 1

print("Number of vowels:", count)
```

```
Enter a string:  a
Number of vowels: 1
```

Challenge Task:

Write a program that prints the Fibonacci series up to n terms using a while loop.

Program:

```
n = int(input("Enter the number of terms: "))
a, b = 0, 1
count = 0

while count < n:
    print(a)
    a, b = b, a + b
    count += 1
```

```
Enter the number of terms:  3
0
1
1
```